MLP Trees

Bagged Expert Ensembles for Tabular Data

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Interim Presentation 2026
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October 2026

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Problem Statement

The Challenge: Why Do Trees Still Win?

Tree models are strong on tabular data because they:

- work directly with individual features
- naturally model threshold-like decisions
- benefit from ensembles of many learners

Our Core Questions

1. Can a neural ensemble genuinely close this gap?
2. Can we do it, without losing full end-to-end differentiability?

Our Approach: Bagged Expert Ensembles (BEE)

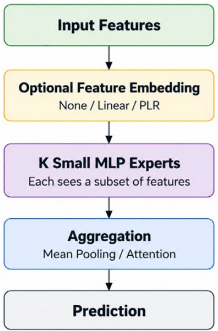
The Pipeline Flow:

Feature bagging: encourage different experts to learn different information

Feature embeddings: process each feature separately before they are mixed

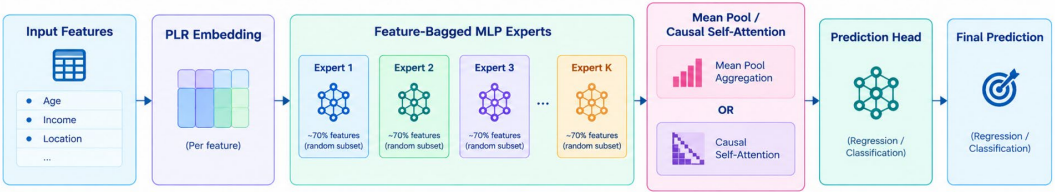
Aggregation: combine the expert representations

Key Early Finding: Mean pooling remained a strong and simple baseline across our experiments.



Architecture

Architecture



Questions We Investigated

Question 1: Are the Experts Learning the Same Thing?

Without Feature Bagging
Expert 1 $\rightarrow$ A B C D E
Expert 2 $\rightarrow$ A B C D E
Expert 3 $\rightarrow$ A B C D E

With Feature Bagging
Expert 1 $\rightarrow$ A B D
Expert 2 $\rightarrow$ B C E
Expert 3 $\rightarrow$ A D E

Each expert sees $\sim$70% of the features.

Verified, Mean off-diagonal cosine similarity: 0.108

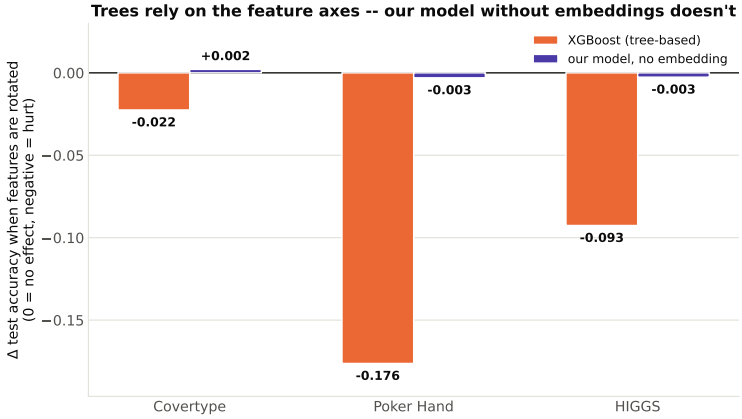
Lower similarity suggests the learned expert representations are not identical.

Question 2: Does the Model Depend on the Original Feature Axes?

What is Rotation Invariance?
In tabular data, features have distinct, independent meanings (e.g., Age vs. Income). Rotation mathematically scrambles these features together. A model is "rotation-invariant" if it treats the distinct data and the scrambled data exactly the same.

The "17-Point" Diagnostic Test – Poker Hand [5]
Rotating input data destroys XGBoost's accuracy (e.g., a 17-point drop) because trees require unmixed axes to make splits. However, it cost our un-embedded MLP essentially nothing. This proved our neural network was natively blurring features together.

Question 2: Does the Model Depend on the Original Feature Axes?

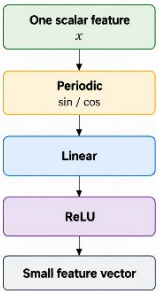


The Fix: PLR tokenizes each feature independently before they ever interact.

PLR: Processing Each Feature Before Mixing

"PLR" = Periodic → Linear → ReLU *(Gorishniy et al., 2022)* [2]

1. **PERIODIC:** Map value into $\sin / \cos$ waves at *learned* frequencies.
2. **LINEAR:** Shrink to a small per-feature vector.
3. **ReLU:** Standard non-linearity.



Why it Works

PLR takes one numerical feature and turns it into a small nonlinear vector before any feature mixing happens.

Question 3: Does Attention Earn Its Place?

Test: 8 benchmark datasets, same experts, same size (∼38k parameters), same training. The only difference is how expert opinions are combined: **simply average them**, or **causal attention**.

Dataset	Averaging	Causal attention	Better
California Housing (RMSE↓)*	0.5404	**0.4985**	attention
Adult (accuracy↑)	**0.8589**	0.8567	averaging
Jannis (accuracy↑)	**0.7195**	0.7097	averaging
Covertype (accuracy↑)	**0.9324**	0.9100	averaging
YearPredictionMSD (RMSE↓)	**8.8293**	8.8472	averaging (within noise)
Higgs Small (accuracy↑)	**0.7215**	0.7173	averaging
Facebook Comments (RMSE↓)	**5.9443**	6.0738	averaging
Santander (accuracy↑)	**0.9147**	0.9131	averaging

Result: averaging wins on **7 of 8** datasets, and attention takes 2–3× longer to train. **So we simply average.**

*These runs were capped at 60 epochs, which cut averaging short on California Housing. In our follow-up test trained to convergence (5 seeds each), averaging wins there too.

Experimental Results

Related Work: Model Comparison (Santander)

Santander - Binary Classification Accuracy (Higher is better)

Model	Accuracy	Source
MLP-PLR	0.9240	Published [2]
XGBoost [7]	0.9235	Our Tuned run
CatBoost [8]	0.9230	Published [2]
BEE Mean-pool + PLR	**0.9227**	Our Un-Tuned run
Published XGBoost	0.9180	Published [2]
Architecture	0.9140	Predecessor
PlainMLP Baseline	0.9112	Our run

BEE performs close to the strongest compared baselines on this dataset.

Results: Headline Wins (YearPredictionMSD & Adult)

YearPredictionMSD - RMSE ↓

Model	RMSE
FT-Transformer (published) [1]	8.8550
XGBoost (published) [1]	8.8190
BEE Mean-Pool, no embedding	8.8293

Adult - Accuracy ↑

Model	Accuracy
FT-Transformer (published) [1]	0.8590
BEE Causal + PLR	0.8608

The Embedding Exception: On YearPredictionMSD, PLR embeddings actively *regressed* performance (8.8293 → 8.8621). This proves PLR is not a universal silver bullet, justifying our modular, switchable design.

The Persistent Gap: Covertype

Covertype (581k rows, 54 features) — the project's clear weak spot, and the target of a dedicated tuning + architecture investigation.

Model	Test Accuracy
TabM (5-model ensemble) [4]	0.9774
TabR (5-model ensemble) [3]	0.9770
Published FT-Transformer [1]	0.9700
Our XGBoost (tuned)	0.9689
BEE Mean-pool — TUNED	**0.9569**

Per-dataset Optuna tuning closed 41% of the gap to XGBoost (0.9324 → 0.9569). **Feature gating** closed *none* of what remained. Checked against the current state of the art (TabR/TabM), not just one 2021-era baseline — the gap is architectural, not a tuning artifact, and not specific to which published model we compare against.

Introduction to Negative Correlation Learning

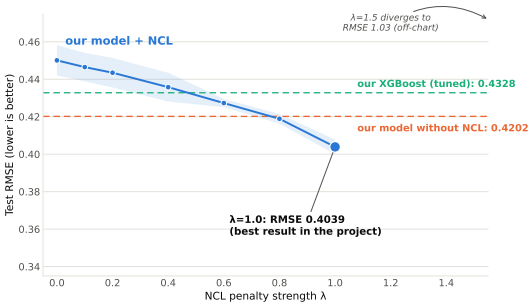
Negative Correlation Learning (NCL)

Idea [6]**:** Instead of forcing diversity architecturally (feature bagging alone), penalize learners for making the *same mistakes*, directly in the loss function.

$$\mathcal{L}_i = \text{MSE}(F_i, y) - \lambda(F_i - \overline{F})^2$$

- **Accuracy** is rewarded (MSE term).
- **Distance** from the ensemble mean is rewarded too (penalty term)—pulling learners apart to cover blind spots.
- Requires moving pooling from embedding-space to output-space, so each learner has its own scalar prediction to penalize.

The Strongest Result: NCL on California Housing



Model	RMSE
TabR (2023, single) [3]	0.4010
BEE λ=1.0	**0.4039**
Without NCL	0.4202
TabM (2025, single) [4]	0.4292
Our XGBoost (tuned)	0.4328
Pub. FT-Transformer [1]	0.4590

Beats our tuned XGBoost, published FT-Transformer, **and TabM (2025)** — within 0.003 of TabR, the closest our model has come to the current state of the art. Not even jointly tuned with λ: the architecture was kept fixed and only λ was varied.

NCL Results: Does It Generalize Beyond California Housing?

Dataset	Task	Without NCL	With NCL	Effect
California Housing (RMSE ↓)	Regression	0.4202	**0.4039** ($\lambda = 1.0$)	✓ helps (beats XGBoost)
Facebook (RMSE ↓)	Regression	5.826	**5.673** ($\lambda = 0.8$)	✓ helps
Higgs Small (Acc ↑)	Classification	0.7245	≈0.51 (all $\lambda > 0$)	× collapses to one class
Santander (Acc ↑)	Classification	0.9218	0.8995 (all $\lambda > 0$)	× predicts majority class

Bottom line: NCL helps on regression (2 of 2) but collapses on classification (0 of 2).

Why classification breaks: the penalty pushes on unbounded logits, so the loss loses its lower bound and training blows up. Fixing this (probability-space NCL) is our first next step.

Conclusion & Next Steps

Scorecard & Limitations

Main Findings

- **Feature Bagging – Useful**
  Different feature subsets encourage expert diversity.
- **Feature Embeddings - Useful, but dataset dependent**
  PLR helps in several cases but not universally.
- **Attention - No consistent advantage**
  Mean pooling is simpler and avoids expert-order dependence.

Main limitation:
Covertype still shows a significant performance gap.

Next Steps

Priority	Proposed Test	Question Answered
1	Probability-Space NCL	Can bounding the penalty fix classification?
2	In depth Analysis Pooling vs. Attention	Can new changes to the Attention architecture outdo simple Mean-Pooling?
3	Per-Dataset Optuna Tuning	What is our architecture's true ceiling?

Datasets Used in Our Experiments

Dataset	Task	Rows	Input Features
California Housing	Regression	**20,640**	**8** numerical
Covertype	Multiclass	**581,012**	**54** numerical
Jannis	Multiclass	**83,733**	**54** numerical
YearPredictionMSD	Regression	**515,345**	**90** numerical
Facebook Comments	Regression	**197,080**	**51** (50 numerical + 1 categorical)
Santander	Binary classification	**200,000**	**200** numerical
Higgs Small	Binary classification	**98,049**	**28** numerical
Adult	Binary classification	**48,842**	**14** (6 numerical + 8 categorical)

Thank You

Questions & Discussion

Appendix: Datasets Used

Dataset	Rows	Features	Task	Description
California Housing	20,640	8	Regression	Median house value per district (1990 US census)
Adult	48,842	108*	Binary	Does a person earn over $50k a year? (US census)
Jannis	83,733	54	4-class	Anonymised benchmark from the ChaLearn AutoML challenge
Higgs Small	98,049	28	Binary	Particle physics: Higgs-boson signal vs. background
Facebook Comments	197,080	131*	Regression	Number of comments a Facebook post will receive
Santander	200,000	200	Binary	Will a bank customer make a transaction? (90% one class)
YearPredictionMSD	515,345	90	Regression	Release year of a song from its audio features
Covertype	581,012	54	7-class	Forest cover type from terrain features
Poker Hand†	1,025,010	10	10-class	Rank of a 5-card poker hand
HIGGS†	11,000,000	28	Binary	Full version of the Higgs-boson physics dataset

The first eight use their exact published train/val/test splits [1, 2], so results compare directly with published numbers.

*After one-hot encoding the categorical columns. †Used only in the rotation/noise/smoothing diagnostics, as 35k-row samples.

References

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[2] Y. Gorishniy, I. Rubachev, A. Babenko. *On Embeddings for Numerical Features in Tabular Deep Learning*. NeurIPS 2022. arXiv:2203.05556. PLR embeddings; Santander results

[3] Y. Gorishniy, I. Rubachev, N. Kartashev, D. Shlenskii, A. Kotelnikov, A. Babenko. *TabR: Tabular Deep Learning Meets Nearest Neighbors*. ICLR 2024. arXiv:2307.14338. Numbers from the official code release, github.com/yandex-research/tabular-dl-tabr.

[4] Y. Gorishniy, A. Kotelnikov, A. Babenko. *TabM: Advancing Tabular Deep Learning with Parameter-Efficient Ensembling*. ICLR 2025. arXiv:2410.24210. Numbers from the official code release, github.com/yandex-research/tabm.

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[7] T. Chen, C. Guestrin. *XGBoost: A Scalable Tree Boosting System*. KDD 2016.

[8] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, A. Gulin. *CatBoost: unbiased boosting with categorical features*. NeurIPS 2018.